LEARNER GUIDE

Bounds in problems

Percentages, Ratio, Growth and Bounds · Lesson 10. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The bounds of an answer tell you how sure you can be. To make a decision (enough? under the limit? will it fit?), check whether every possible value gives the same answer: if not, say you can't be sure. To give an answer to a **suitable degree of accuracy**, round the lower and upper bounds: give the answer to the most accurate degree at which both round to the same value.

What you are learning to notice

- Find both bounds of the answer before you decide anything.
- “Can't be sure” is an honest answer when the interval reaches both sides.
- For “enough”, plan for the upper bound: the most you could need.
- Round both bounds: where they agree, you can be sure.
- Measuring more precisely narrows the interval, so you can be surer.

Moving through it, one step at a time

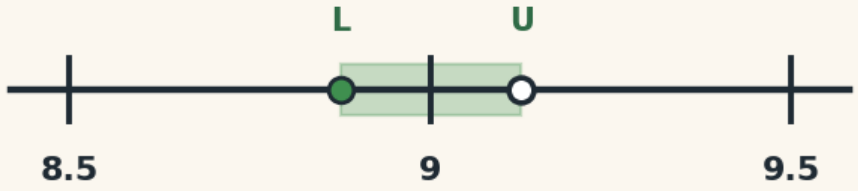
Write each measurement's bounds. Find the upper and lower bounds of the answer (Lesson 9). For a decision, compare the whole interval with the limit. For the accuracy, round both bounds to 1 significant figure, then 2, then 3, and give the answer to the most accurate degree at which they match.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A suitable degree of accuracy

A cyclist covers 360 m in 40 s, both to the nearest whole number. Give the speed to a suitable degree of accuracy. (On the line, *L* and *U* mark the lower and upper bounds of the speed.)



1. Lower bound: smallest over largest $\frac{359.5}{40.5} = \frac{719}{81} = 8.88$ to 2 d.p.

2. Upper bound: largest over smallest $\frac{360.5}{39.5} = \frac{721}{79} = 9.13$ to 2 d.p.

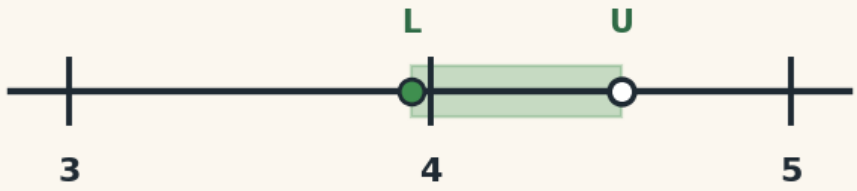
3. To 2 s.f. 8.9 and 9.1: different

4. To 1 s.f. 9 and 9: the same

9 m/s, to 1 significant figure. (Both bounds lie between 8.5 and 9.5, so every possible speed rounds to 9.)

Example 2 — Is it enough?

A wall is 38 m^2 and one tin of paint covers 9 m^2, both to the nearest m^2. How many tins are sure to be enough? (*L* and *U* mark the fewest and the most tins you could need.)



1. The most tins you could need	$\frac{38.5}{8.5} = 4.53$ to 2 d.p.
2. The fewest you could need	$\frac{37.5}{9.5} = 3.95$ to 2 d.p.
3. Decide	4 tins might not be enough; 5 always are

5 tins are sure to be enough.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (calculating with bounds (Lesson 9) and rounding to significant figures), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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